Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Full-stack engineer who **built AgentProbe end-to-end** - the exact platform Scale's Enterprise AI team builds: APIs for knowledge retrieval (RAG + ChromaDB), LLM inference routing (Anthropic, OpenAI, Groq, Ollama), and a **5-metric evaluation engine** with CI/CD regression gates - all production-grade, open-source. **8+ years Python**; full-stack delivery across 120+ Agile sprints at Entrata (100K+ MAU, 99.5% uptime, 1.6M financial records); TypeScript/JavaScript across 20+ modules. AWS (S3, EC2, RDS, Glue) in production at scale. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

Python (8+ years production) | LLM Integration & Inference APIs | Knowledge Retrieval (RAG + ChromaDB)

LLM Evaluation Framework | Multi-Agent Orchestration (LangGraph) | TypeScript / JavaScript / React

FastAPI / REST Platform Layer | AWS (S3, EC2, RDS, Glue) | Docker / Microservices | CI/CD (GitHub Actions)

Production at Scale (1M+ records, 100K+ MAU) | SQL / PostgreSQL / MySQL

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Built and delivered **20+ full-stack modules** (Python, TypeScript/JavaScript, PostgreSQL) across **120+ Agile sprints** for **100K+ monthly active users** - owned large product areas end-to-end from scoping through production.
- Applied **Python, PyTorch, and OpenAI API** to automate data transformation across **1.6M financial records** - production AI on regulated data with output validation and quality monitoring.
- Led Docker-based containerization for **30%+ of microservices**, cutting execution time by **50%**; improved production query performance by **30%** across 200+ SQL queries; maintained **99.5% uptime** for 6+ months.
- Migrated client data to **AWS S3** with Libsodium-encrypted pipelines - **100% PHI compliance, zero security incidents** across 100K+ users; managed EC2, RDS, and Glue.
- Built **5+ internal platform tools** adopted across squads, reducing resolution time by **40%**; drove secure coding standards and peer code review practices across all releases.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Mentored 30+ students in Python and database systems; designed supplementary exercises and drove iterative quality improvement across the course.

PROJECTS

AgentProbe: Enterprise GenAI Platform - Knowledge Retrieval, Inference, Evaluation APIs OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

Built Scale's enterprise AI platform equivalent - APIs for knowledge retrieval, LLM inference, and evaluation - from greenfield to open-source production. **Knowledge retrieval API**: ChromaDB + embeddings pipeline with retrieval quality scoring across multi-step agent runs - the same RAG API pattern Scale's enterprise customers use. **Inference API**: multi-provider LLM routing across Anthropic, OpenAI, Groq, Ollama - dynamic selection by task type, latency, and capability at the API layer. **Evaluation API**: 5-metric LLM-as-judge engine (factual accuracy, hallucination, retrieval, completeness, compliance) with per-run regression tracking and CI/CD gates - shared eval tooling any team can call. **Multi-agent runtime**: 3-agent LangGraph system (Orchestrator, RAG Tracer, Auto Fix) with tool routing and inter-agent state handoffs. **Platform layer**: 15-endpoint FastAPI backend + React observability dashboard - the shared abstraction product teams build on top of.
github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR: Domain-Specific NLP Platform - LLMs + FHIR APIs

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings

NLP platform routing natural language queries to domain-specific FHIR APIs via LLMs, BERT, and embeddings - HIPAA-compliant RBAC. **~60% retrieval time reduction.** Built for a regulated enterprise domain with strict compliance requirements.
github.com/venkatalakshmiparimala/AI_on_FHIR

CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection → evaluation → deployment. **85%+ accuracy.**

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI & Platform: LangChain, LangGraph, Multi-Agent Orchestration, LLM APIs (Anthropic, OpenAI, Groq, Ollama), RAG Pipelines, ChromaDB, Vector Search, LLM-as-Judge Evaluation, CI/CD Eval Gates, Prompt Engineering, Context Engineering

Backend & Infra: FastAPI, Flask, React, PostgreSQL, MySQL, Docker, AWS (S3, EC2, RDS, Glue), Azure, CI/CD (GitHub Actions), RabbitMQ, Microservices

ML & Data: PyTorch, scikit-learn, BERT, spaCy, Hugging Face, NumPy, Pandas, Feature Engineering, Statistical Modeling

Process: TDD/BDD, Integration Testing, Open-Source Development, Git, JIRA, Agile/Scrum